



REAL-TIME TRAFFIC SIMULATION WITH SUMO PLANNING & TRACKING

Slides made by **Duc Thien Nguyen**

PROJECT TIMELINE

TEAM MEMBER
MARCO
LAZAR
DUC THIEN
HOANG BACH
MINH PHU



OUR MAIN ROLES

	ROLE	RESPONSIBILITIES
MARKO	SUMO INTERGRATION TEAM LEADER	SUMO CONNECTION VIA LIBTRACI API , WRAPPER
LAZAR	GUI LEAD	Map visualization, camera controls (zoom/pan/rotate), UI panels for injection, Ttraffic Light control, filters, charts.
DUC THIEN	DOCUMENTATION AND TESTING	user manual/screenshots, UML diagrams, milestone PDF assembly, declarations of authorship.
HOANG BACH	DATA ANALYSIS	Traffic statistics computation, real-time charts, CSV and PDF report generation, file and stream handling
MINH PHU	CORE SIMULATION	Simulation loop implementation, SUMO stepping logic, thread management, synchronization between simulation and GUI

MILESTONE ONE

Due on : 27/11/2025

Project: New campaign

Task	Status	Priority	Done Date	Owner	Lecturer's Comment
SUMO connection demo and libtraci wrapper	✓ Completed ▾	● High ▾	27 Nov 2025	Marko	Add more features in milestone 2
GUI Mockups	✓ Completed ▾	● High ▾	27 Nov 2025	Lazar	Make GUI and Visualize
Sumo networking creating and testing	✓ Completed ▾	● High ▾	27 Nov 2025	Hoang Bach	Absent in Milestion 1
Sumo networking creating and testing	✓ Completed ▾	● High ▾	27 Nov 2025	Minh Phu	Absent in Milestion 1
Report and architecture diagram	✓ Completed ▾	● High ▾	27 Nov 2025	Duc Thien	Fix the diagram and make the timeline

MILESTONE TWO

Due on : 14/12/2025

Project: New campaign

Task	Status	Priority	Done Date	Owner	Notes
Sumo Wrapper, check API key	✓ Completed ▾	● High ▾	14 Dec 2025	Marko	
GUI (Center control)	✓ Completed ▾	● High ▾	📅	Lazar	Not response
SUMO Stress Test	✓ Completed ▾	● High ▾	📅	Hoang Bach	Not response
Application (Map visualization, Vehicle injection. traffic light	✓ Completed ▾	● High ▾	09 Dec 2025	Minh Phu	
User Guide Draft, Testing, Report	✓ Completed ▾	● High ▾	14 Dec 2025	Duc Thien	GUI draft (using Swing , look at JavaFX later)